RAVEL: A Spatial Coding Harness for Dynamic Relational Reasoning

Anonymous CVPR submission

Abstract

Understanding how the physical world changes is central to embodied intelligence. A key part of this understanding is reasoning about how actions change or preserve spatial relations among objects. Yet vision-language models still struggle to maintain consistent judgments across multiple relations as a scene evolves. We introduce RAVEL, a training-free spatial coding harness that represents objects, actions, relations, and evidence in a persistent, executable world model. It couples state updates with rules to track evidence validity, propagate geometry, and evaluate relations across actions. A frozen vision-language model supplies unresolved judgments, which the harness combines with action participation to distinguish change from preservation within and outside an interaction's scope. To test both changed and preserved relations across actions, we introduce an indoor benchmark of 12,000 questions (4,000 test) spanning observed updates, hidden outcomes, constrained edits, and trajectories. Full RAVEL improves all evaluated backends, raising Astra's whole-question accuracy from 62.70% to 81.20% with additional geometry and state-wise calls. Under matched access and schedules, shared representation and execution improve both evaluated backends, with larger gains for Qwen3-VL-4B, which also benefits more from compulsory binding. These results support executable spatial representations for maintaining applicable premises and improving joint relational answers across actions.

Project page: olicy666.github.io/askdhfgk/

1. Introduction

Understanding an action requires reasoning about the relations it changes and preserves. For example, a collision can change contact while preserving support and relations among untouched objects. A correct account must establish relation endpoints and distinguish preservation within and outside the interaction's scope. We call this *selective spatial relation reasoning*.

Prior work grounds spatial judgments in 3D information [4, 6], executes code for reasoning and agent actions [10, 27], and uses scene graphs for task planning [24].

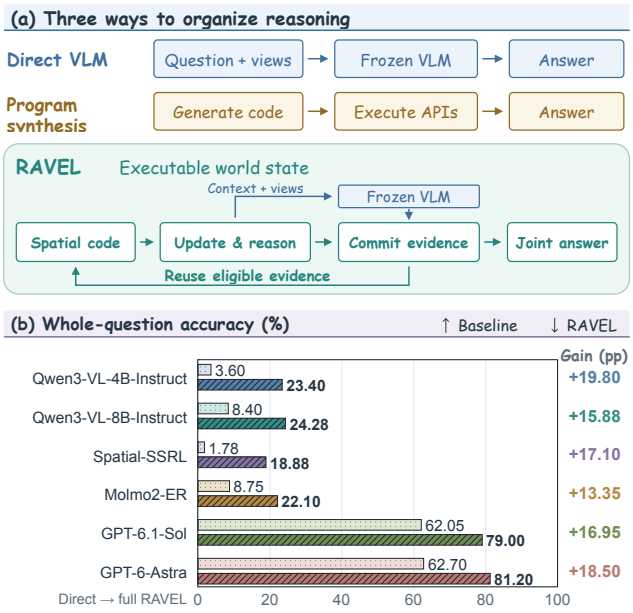


Figure 1. **Spatial coding across actions.** (a) Direct prediction and program synthesis [11, 26] contrast with RAVEL's persistent, executable world state. A frozen VLM supplies unresolved judgments, and committed evidence remains reusable. (b) Whole-question accuracy across six backends (%); gains are percentage-point differences. Upper, dotted bars show the Direct baseline; lower, hatched bars show full RAVEL with additional initial geometry and state-wise calls; matched controls appear in Tab. 2.

We bring relation truth and action participation into a joint evaluation: detecting a collision is insufficient if the answer loses a preserved support relation. Our 12,000-question benchmark tests observed and hidden outcomes, constrained edits, and consecutive actions on a 4,000-question test set. Preservation includes false relations and constrains admissible edits.

The central inference problem is deciding which premises still establish a relation after an action. A past observation may no longer apply, and queries about the same object need different evidence: direction uses centers, while support requires surfaces. A prescribed translation updates a known center; free motion invalidates it as a premise for the final pose. This difficulty is visible in all three local Di-

rect backends scoring 0% Whole on T3 and T5. Meanwhile, 76.5% of tested updates are preservation, so predicting no change can conceal incorrect endpoints.

We introduce Relation-Aware Validity and Evidence Linking (RAVEL), a harness that represents spatial reasoning as executable operations over a persistent world state (Fig. 1). Objects, attributes, actions, queries, and evidence share a code-based representation with update and evaluation rules. Code connects an action to what the system can compute, retain, or observe next: translation preserves a computable center, while release can return a relation to visual judgment. This representation is a working interface between the harness and the frozen model. The harness updates supported attributes and evaluates relations; the model supplies unresolved observations; committed findings retain their sources across states. A fixed interpreter composes typed findings into joint answers and checks candidate goals and preservation constraints. Optional binding switches require the model to adopt computed findings (P) or eligible carried-over findings (L); the underlying representation and execution remain active when binding is disabled.

All five backends improve on the full test set, with additional geometry and state-wise calls. Matched controls show that both tested backends benefit from shared representation and execution; compulsory binding provides a larger additional benefit for Qwen3-VL-4B. The analysis distinguishes evidence that still applies from evidence that is correct, explaining both successful preservation and persistent errors. Our contributions are:

- A spatial coding harness coupling a persistent world state to executable update, validity, evaluation, and composition rules with optional evidence binding.
- A benchmark jointly testing relation truth and participation across observed updates, hidden outcomes, constrained edits, and trajectories.
- Five-backend evaluation and matched interventions separating the joint representation-and-execution benefit from binding and propagation effects.

2. Related Work

Spatial representations and benchmarks. CLEVR [19] tests compositional reasoning; SpatialSense [29] reduces language and image-space shortcuts. SpatialVLM [4] and SpatialRGPT [6] ground spatial judgments in 3D information, while VSI-Bench [28] tests spatial understanding from video. These evaluations complement persistent scene representations: 3D Dynamic Scene Graphs [25] organize objects, places, and moving agents hierarchically to support planning and decision-making. RAVEL couples a persistent spatial representation to action-dependent updates and evidence validity.

Dynamic relations and physical reasoning. Action Genome [18] and STTran [8] model changing relational structure in videos. CLEVRER [33] evaluates causal reasoning about collisions, and PHYRE [2] evaluates solving physical puzzles. Physion [3] tests physical prediction, while PhysBench [7] assesses physical world understanding. EmbodiedBench [30] extends agent evaluation from high-level tasks to low-level navigation and manipulation. Our evaluation jointly scores relation endpoints and action participation, separating preservation during interaction from preservation outside its scope.

Executable visual and spatial reasoning. Neural program execution [20], NS-VQA [32], and NS3D [13] separate visual grounding from compositional reasoning. PAL [10] delegates generated reasoning programs to an interpreter; CodeAct [27] uses executable code as an agent action interface. VISPROG [11], ViperGPT [26], and VADAR [22] generate programs or APIs for visual queries. SceneCraft [14] produces Blender scene code, while VoxPoser [15] composes 3D value maps through code for trajectory planning. SpaceTools [5] learns spatial tool coordination through reinforcement learning. RAVEL executes a fixed interpreter over persistent spatial code, linking each action to applicable evidence and joint answers across states.

Embodied planning and visual memory. ReAct [31] interleaves reasoning and actions; Inner Monologue [16] uses environmental feedback for planning. SayCan [17] grounds skill selection in learned affordances; SayPlan [24] searches hierarchical scene graphs and revises plans using simulator feedback. For temporal evidence, VideoAgent [9] stores event descriptions and object tracking states in structured memory for tool-based querying; VISTA [12] retains raw visual history for active inspection. RAVEL maintains query-specific evidence across a supplied action sequence: it propagates supported attributes, retains eligible findings, and requests unresolved judgments under a fixed schedule.

3. Benchmark: Selective Relation Reasoning

3.1. Relation Truth and Participation

Relation truth alone cannot distinguish support preserved during motion from preservation outside an action's scope. We therefore evaluate truth together with participation. An episode pairs states $s_0, \ldots, s_{L-1}$ with actions $a_0, \ldots, a_{L-2}$ and queries $\mathcal{Q}$. Objects $\{A, B, C, D\}$ have persistent red/blue/green/purple identities and include controlled, free, and fixed participants. Each state has two views, subject to task permissions. Relation values $v_{t,q} \in \{\text{true}, \text{false}\} \cup \{\perp\}$ are true, false, or unknown at a threshold boundary or under insufficient evidence.

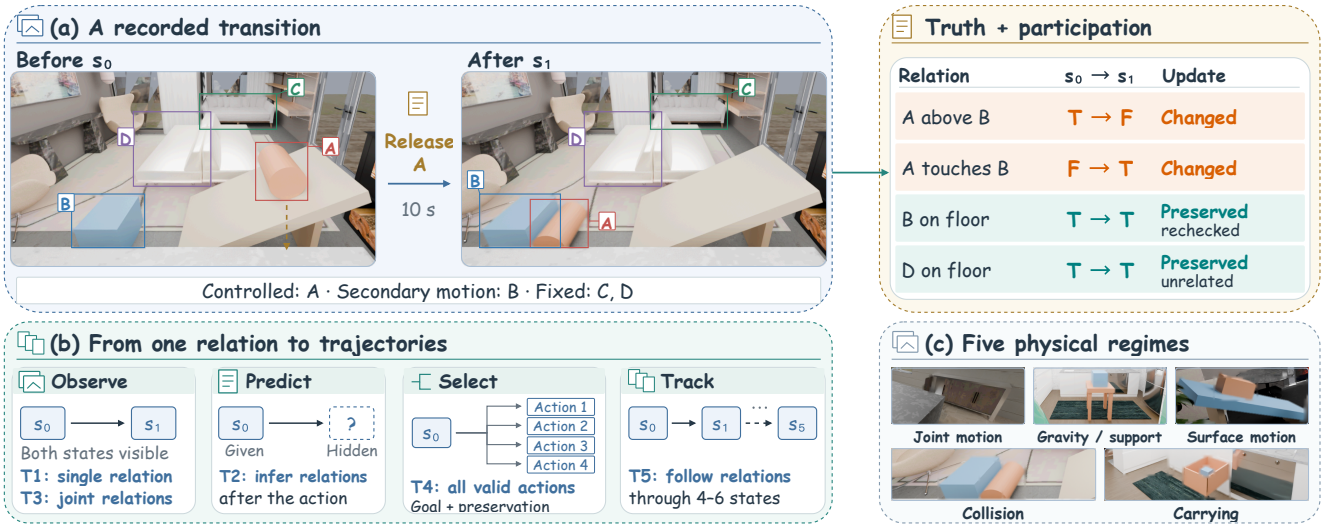


Figure 2. **RAVEL benchmark overview. (a)** A recorded collision connects a release action to before/after views and relational annotations; four of seven queries and one of two views per state are shown. Thin boxes identify objects; the dashed arrow indicates downward release. Orange annotations denote change and teal preservation, distinguishing action participants (rechecked) from objects outside the action (unrelated). **(b)** Five tasks cover observed relations, hidden outcomes, action selection under goals and preservation constraints, and trajectories. T4 candidates independently start from s_0; T5 spans four to six states. **(c)** Recorded scene crops illustrate all five physical regimes. The benchmark contains 12,000 questions, including 4,000 test questions equally divided among five tasks.

The seven predicates—direction, proximity, support, contact, containment, opening, and joint attachment—use center, surface, interior, or joint geometry. Distances use meters in the fixed initial-view-1 ground frame, with vertical z. Direction and proximity use centers; `on_surface` denotes bottom contact or proximity to a named upward surface, not force equilibrium.

Let $\mathcal{D}(q)$ contain the query's named objects and $\mathcal{A}_t$ the directly controlled or secondarily moving participants. An update combines endpoint truth and participation:

$$u_{t,q} = \begin{cases} \text{unknown}, & \bot \in \{v_{t,q}, v_{t+1,q}\}, \\ \text{changed}, & v_{t,q} \neq v_{t+1,q}, \\ \text{rechecked_preserved}, & \mathcal{D}(q) \cap \mathcal{A}_t \neq \varnothing, \\ \text{unrelated_preserved}, & \text{otherwise}. \end{cases} \tag{1}$$

Here $u_{t,q}$ is the update label between states t and $t+1$; $\bot$ marks an unknown endpoint. The last two cases require known, equal values, including false-to-false preservation. Thus B–floor can remain true while B moves, making it rechecked; fixed C–D direction is unrelated. Ground-truth participation describes recorded execution scope, rather than counterfactual causal effects.

3.2. Tasks, Construction, and Evidence Access

The dataset contains 12,000 questions from HSSD [21] and Infinigen Indoors [23] scenes; the test set has 4,000 questions, 800 per task, from 100 source groups. Tasks cover joint motion, gravity/support, surface motion, collision, and carrying. Groups share layouts or assets and define statistical dependence.

T1/T3 score observed endpoints and three-/four-way updates for single/multiple relations; T1 merges preservation types. T2 predicts future relations from initial views and actions. T4 selects all independent edits satisfying every goal and preservation constraint. T5 tracks endpoints and updates across four to six states. T2 futures and T4 candidate outcomes remain hidden; other tasks expose the relevant states.

Construction fixes the task, scene, participants, action parameters, metric scale, and question frame, and declares object mobility and physical properties. Blender executes prescribed articulation and Bullet dynamics; saved transforms and interaction records determine relation endpoints and participation. Each T4 candidate resets to the same initial state. Two views are rendered per state. Automated checks cover all 12,000 questions, verifying execution, file completeness, physical conditions, annotation consistency, and input permissions. Two reviewers each inspected 250 of 500 randomly sampled test questions spanning all tasks and physical categories, checking object references, action plausibility, and image–answer consistency. Failed jobs are adjusted or resampled and rerun independently of model performance. Inputs contain permitted images, actions, declarations, queries, and schemas; hidden outcomes and mesh-level scoring evidence remain private.

The test set contributes 77,303 answer items, including 27,664 T3/T5 updates; 76.5% of these updates are

preservation. This motivates evaluating change recall and preservation-specific errors.

4. RAVEL: A Spatial Coding Harness

JSON provides structured interfaces for exchanging objects, attributes, relations, and actions. RAVEL uses spatial code to define executable operations over these records: state updates, dependency checks, geometry propagation, and answer composition. A fixed interpreter applies these operations to a persistent world state. Translation updates a valid center; free motion invalidates it and redirects dependent queries to model judgment. JSON remains the input/output format, while code governs how spatial evidence is computed, retained, and combined across calls. The frozen model supplies unresolved observations from the state and permitted views.

The harness has three stages: update the spatial code, organize applicable evidence, and commit findings before composing answers. All stages remain active in C00 with binding disabled. Optional P/L switches enforce computed/eligible carried-over findings; full RAVEL (C11) enables both.

4.1. Stage 1: Construct and Update Spatial Code

RAVEL compiles public recipes, actions, and queries into spatial code (Fig. 3). `SceneSpec` describes objects, geometry, and valid attributes; `ActionProgram` describes operations; queries and findings record dependencies and provenance. The model reads this representation, and a fixed interpreter executes it.

Interpreting actions reconstructs current attributes from the recipe and history. Fixed objects retain geometry; prescribed translation updates a valid center as $c_{t+1}(o) = c_t(o) + \Delta_t$. Rotation and free evolution of unheld objects invalidate their centers; joint operations update declared opening coordinates. Initial geometry remains historical when `current_geometry` is absent. Initial centers are computed from the objects' 3D axis-aligned bounding boxes (AABBs), transformed into the question's coordinate frame, and supplied equally to matched Direct. The interpreter does not simulate free-body trajectories.

4.2. Stage 2: Compute and Organize Relation Evidence

Each query uses a supported computation, an applicable prior finding, or a fresh judgment. These choices follow relational dependencies, so motion need not affect all queries alike. Evidence is computed or retained before the P/L switches determine whether its adoption is compulsory.

Computed evidence. A relation supported by valid geometry should not depend on a new visual estimate. Direction and proximity execute when both current centers are valid and stored error bounds permit a threshold decision. Declared or prescribed joint coordinates establish opening; authored joint topology establishes attachment. The computed result becomes an explicit finding in the model-facing representation. With program binding P enabled, the structured output requires its adoption, ahead of carried-over model evidence. With P disabled, the same valid computation remains available as advisory evidence. After A's release, C–D direction remains computable; A and B lose authoritative centers.

Carried-over evidence. An observation about fixed objects need not be estimated again after an unrelated action. A known finding is retained when all named operands are fixed. It retains its original state and evidence basis, as with D–floor support in the collision example. The fixed-dependency check establishes applicability, while the earlier observation can still be wrong. The L switch controls compulsory adoption of that eligible observation; disabling L retains its content and provenance as advisory evidence.

Fresh evidence. Queries without a supported computation or eligible prior observation require a model judgment. The model resolves remaining relations using permitted images, spatial code, and predicate-specific vocabularies. Contact, support, and containment require surface or interior evidence beyond centers and bounding boxes. The harness projects findings to true, false, or null. For participation, direct targets are controlled and fixed objects still; the model judges secondary motion of other objects using a 0.025 m center-displacement threshold. Shared participant status then enters every relevant update.

4.3. Stage 3: Commit Findings and Compose Answers

Each call exposes spatial code and a typed schema. P/L constrain established findings to singleton choices; disabling them opens the corresponding vocabularies while preserving state updates, computation, history, and composition. The harness validates and commits model judgments with provenance for subsequent calls; the model supplies relational evidence without updating numerical geometry.

Endpoints establish change, and shared participation distinguishes preservation types through Eq. (1); T1 merges those types.

In the collision example, A's `current_geometry` becomes `None`; its initial geometry remains historical. The model detects a true-to-false change in A–B height ordering. B–floor support stays true, but B's secondary motion makes it rechecked-preserved; fixed D–floor support is unrelated-preserved. These updates combine query-specific evidence with shared participation rather than assigning one change label to the scene.

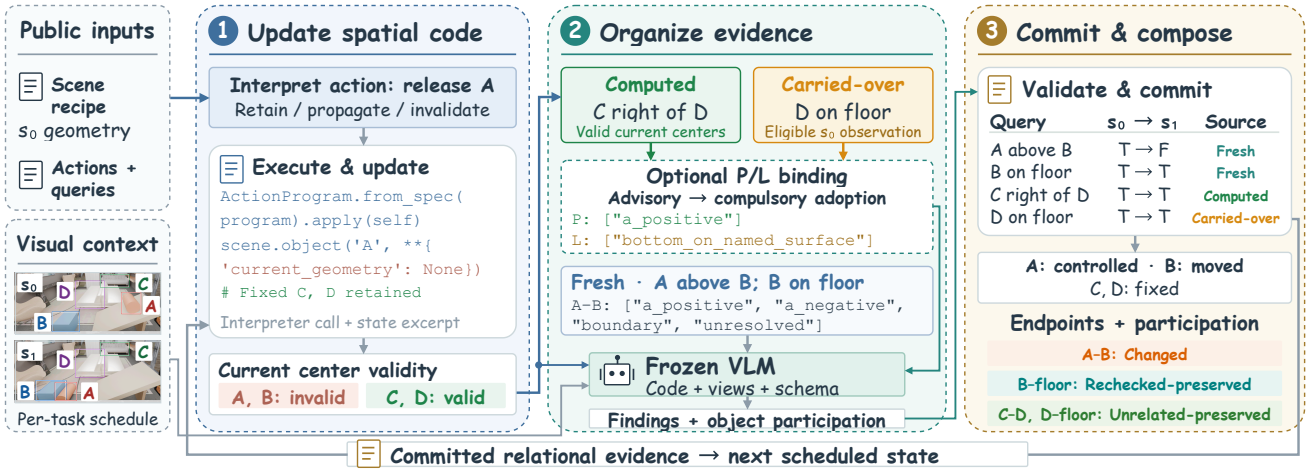


Figure 3. **Shared harness execution with optional evidence binding.** A collision example connects an interpreter call to updated spatial code, evidence selection, and joint answers. C00 uses advisory evidence; C11 enables P/L binding, with computed findings taking precedence over carried-over findings. Singleton choices enforce established findings, while Fresh queries retain open vocabularies (direction shown). Code excerpts omit other fields.

For T4, let $\mathcal{G}$ contain goal query–value pairs, $\mathcal{K}$ preservation queries, and $\hat{v}_{c,q}$ the prediction under independent candidate c:

$$G(c) = \bigwedge_{(q,b)\in\mathcal{G}} [\hat{v}_{c,q} = b],$$
$$K(c) = \bigwedge_{q\in\mathcal{K}} [(\hat{v}_{0,q} \neq \bot) \wedge (\hat{v}_{c,q} = \hat{v}_{0,q})], \quad (2)$$
$$\widehat{\mathcal{C}} = \{c \in \mathcal{C} : G(c) \wedge K(c)\}.$$

Acceptance requires predicted findings to satisfy every goal and preserve each specified known value, including false values. A successful goal cannot mask a violated constraint; correctness still depends on the findings.

5. Experiments

5.1. Setup

We evaluate on the 4,000-question test set, with 800 questions per task, using Qwen3-VL-4B-Instruct [1], Spatial-SSRL-4B, Molmo2-ER, and API backends Sol (`gpt-6.1-sol`) and Astra (`gpt-6-astra`), both at medium reasoning effort. Direct receives the question, permitted images, common semantics, and final schema in one call. RAVEL additionally receives public initial geometry and declarations and uses state-wise inference. Both use constrained structured output without answer repair; all 40,000 main answers have valid format.

Whole-question accuracy requires every semantic item to be correct; Items micro-averages individual decisions. T4 Whole requires the exact admissible set. For T3/T5, changed F1 measures change detection, while joint accuracy requires both endpoints and the update label. T5 further distinguishes state-vector from complete-trajectory correctness.

Implementation. T1/T3 call once per endpoint; T5 calls once per state, using current and adjacent previous two-view images with persistent actions and findings. T2/T4 each use one initial-view call for a future state/four independent candidate branches. Hidden endpoints and ground-truth participation are unavailable; free-body outcomes remain model predictions.

5.2. Main Results

RAVEL improves whole-question accuracy across the five evaluated backends (Tab. 1). For example, Astra improves from 62.70% to 81.20%, with gains across all five tasks. The local backends also improve, but their much higher item accuracy than whole-question accuracy reveals a remaining difficulty: individual relations can be correct while their joint answer is incomplete. The full comparison adds initial declarations and state-wise calls alongside the harness. We next hold these inputs and schedules fixed to examine the role of spatial representation and execution.

5.3. Matched Controls and Component Effects

We sample 500 of the 4,000 test questions to compare matched Direct with harness variants and assess shared representation and execution, P/L binding, and selective propagation. The subset contains 100 questions per task and 20 per task–physics cell, spanning all 100 source groups. Within each backend, controls share initial geometry and physical declarations, permitted images, call schedules, and per-call output limits. Matched Direct stores and assem-

Table 1. **Full-cohort results (4,000 questions; 77,303 answer items).** Metrics are percentages; – denotes unavailable or inapplicable entries; ΔWhole is RAVEL minus Direct in points. Bold marks the better Whole, Items, T5, and F1 within each backend. Joint accuracy requires both endpoints and the T3/T5 update. RAVEL adds declarations and uses 8,995 calls versus 4,000; matched controls use common information and call schedules (Tab. 2).

		Overall			Whole by task					Changed	Joint endpoints + update		
Backend	Method	Whole	Items	ΔWhole	T1	T2	T3	T4	T5	F1	Changed	Recheck.	Unrel.
Qwen3-VL-4B-Instruct	Direct	3.60	44.66	–	14.38	2.25	0.00	1.83	0.00	18.85	5.18	16.94	38.50
	RAVEL	**23.40**	**78.92**	+19.80	67.75	10.50	12.60	23.17	**5.63**	**62.40**	50.36	62.68	87.98
Qwen3-VL-8B-Instruct	Direct	–	–	–	–	–	–	–	–	–	–	–	–
	RAVEL	–	–	–	–	–	–	–	–	–	–	–	–
Qwen3-VL-32B-Instruct	Direct	–	–	–	–	–	–	–	–	–	–	–	–
	RAVEL	–	–	–	–	–	–	–	–	–	–	–	–
Spatial-SSRL-Qwen3VL-4B	Direct	1.78	37.59	–	8.00	0.13	0.00	1.00	0.00	12.14	2.04	7.96	39.52
	RAVEL	**18.88**	**75.74**	+17.10	62.00	6.75	7.30	17.67	**3.25**	**56.06**	43.28	57.56	87.39
Molmo2-ER	Direct	8.75	47.45	–	39.50	2.75	0.00	2.00	0.00	38.03	0.49	39.84	0.00
	RAVEL	**22.10**	**74.67**	+13.35	65.50	11.13	9.20	22.50	**5.50**	**53.13**	35.33	62.34	78.84
MiniCPM-V4.5	Direct	–	–	–	–	–	–	–	–	–	–	–	–
	RAVEL	–	–	–	–	–	–	–	–	–	–	–	–
DeepSeek-4.1-Flash	Direct	–	–	–	–	–	–	–	–	–	–	–	–
	RAVEL	–	–	–	–	–	–	–	–	–	–	–	–
GPT-6-Luna	Direct	–	–	–	–	–	–	–	–	–	–	–	–
	RAVEL	–	–	–	–	–	–	–	–	–	–	–	–
GPT-6.1-Sol	Direct	62.05	92.76	–	88.13	55.38	55.20	64.00	49.75	88.13	81.63	93.08	94.84
	RAVEL	**79.00**	**96.64**	+16.95	96.00	69.00	77.80	80.83	**72.13**	**95.49**	93.43	96.81	97.64
GPT-6-Astra	Direct	62.70	92.83	–	87.50	57.38	54.30	66.50	50.88	88.25	81.60	93.73	94.30
	RAVEL	**81.20**	**96.96**	+18.50	95.75	73.75	79.10	86.50	**72.75**	**95.71**	94.30	96.78	97.54

Table 2. **Matched controls (500 test questions).** P/L bind computed/retained findings; all C rows retain the shared harness. Cons. invalidates non-fixed centers after each action. Bold: column best. Differences are in percentage points and rounded separately.

			Qwen3-VL-4B			Astra		
Configuration	P	L	Whole	Items	F1	Whole	Items	F1
Direct	–	–	6.20	52.64	23.65	78.00	96.14	94.57
C00: Binding off	0	0	21.60	78.56	62.15	80.00	**96.73**	95.53
C10: P only	1	0	23.20	79.21	63.82	78.40	96.53	94.66
C01: L only	0	1	21.40	78.86	62.64	79.40	96.59	95.03
C11: Full RAVEL	1	1	**24.40**	**80.03**	**65.50**	**80.20**	96.59	**97.51**
Cons. geometry	1	1	20.40	78.09	57.45	79.20	96.49	94.71
ΔWhole (C11−Direct)				+18.20			+2.20	
ΔF1 (C11−Direct)				+41.86			+2.94	

bles state-wise predictions from raw declarations; RAVEL adds derived state, relation evaluation, typed findings, and semantic composition.

Effect of the harness under matched access. Under matched information and schedules, Qwen3-VL-4B improves from 6.20% to 24.40% Whole, while Astra improves more modestly from a stronger baseline (Tab. 2). Both also improve changed F1. The larger Qwen3-VL-4B benefit therefore persists when access is matched: how the evidence is used matters alongside its availability.

Representation and execution without compulsory binding. C00 disables P/L binding while retaining structured state, validity tracking, geometry propagation, typed evidence, and answer composition. Even with binding disabled, Qwen3-VL-4B reaches 21.60% Whole, accounting for most of its improvement over matched Direct (Tab. 2). Astra likewise retains most of its matched Whole gain in this configuration. C00 jointly evaluates state updates, relation measurements, attributed history, and answer composition, including prompt and output organization, while allowing the model to disagree with supplied findings.

Incremental effect of compulsory binding. The P/L factorial comparison keeps that harness active and varies whether the output must adopt its computed and eligible carried-over findings. Enabling both further improves Qwen3-VL-4B to 24.40% Whole (Tab. 2). This additional gain measures the value of requiring the model to adopt evidence already supplied by the shared harness.

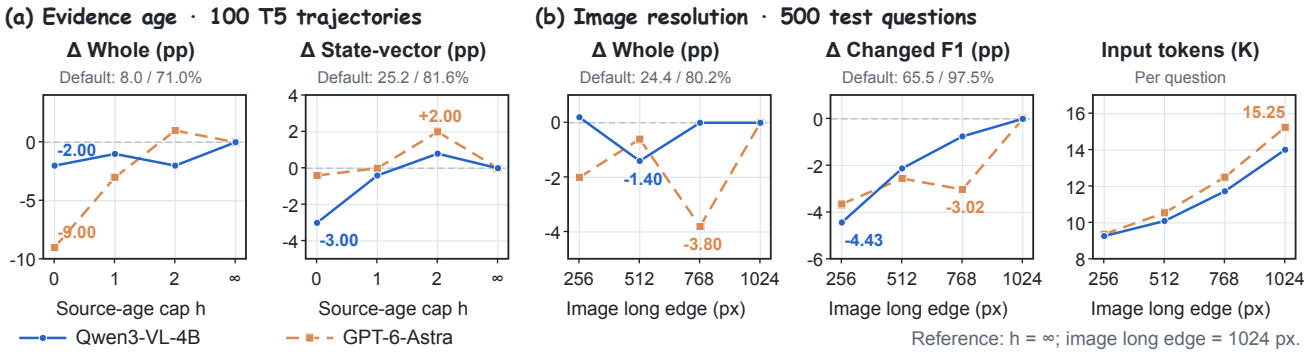


Figure 4. **Evidence retention and visual detail affect different decisions.** (a) Source-age caps on 100 T5 trajectories. (b) Image resolution on 500 test questions. Quality changes are relative to unlimited history or 1,024 pixels; default scores are Qwen3-VL-4B/Astra. Tokens are per-question means.

The two binding mechanisms interact: enforcing carried-over evidence alone slightly lowers Qwen3-VL-4B Whole, whereas combining it with computed-evidence enforcement improves performance. Astra already adopts all harness-provided computations without compulsory binding, with most of its matched gain realized in C00. Joint binding adds a small Whole gain in this setting. Its additional value depends on how the backend uses the supplied evidence; the shared representation and execution remain active in both configurations.

5.4. Design Choices

Effect of selective geometry propagation. A known translation updates a moving object's center within the representation. We test this distinction by conservatively invalidating every non-fixed center after each action. Restoring selective propagation raises Qwen3-VL-4B Whole from 20.40% to 24.40% and also improves changed F1 (Tab. 2). Astra improves on both metrics, with a smaller Whole gain. Discarding valid premises returns computable relations to model judgment. Propagation determines which premises remain available; binding determines whether available findings must be adopted. These contrasts use different reference configurations, so their gains are not additive.

Evidence retention. On 100 matched T5 trajectories, we cap original-source age at $h \in \{0, 1, 2, \infty\}$ while retaining images, actions, initial declarations, and computation; reuse does not reset age. Shorter history does not consistently improve Whole (Fig. 4a). Astra performs best with an intermediate cap, while Qwen3-VL-4B favors unlimited retention. Age cuts remove correct and incorrect findings together, so recency alone cannot identify which evidence to discard.

Visual evidence. Reducing image resolution saves tokens, but nearly stable Whole can conceal a loss in change detection (Fig. 4b). At 256 pixels, Qwen3-VL-4B's changed F1 falls from the default 65.50% to 61.07%. Supported computations remain available at lower resolution, while unresolved judgments need image detail. Under paired lighting/material changes to the same episodes, both backends retain their advantage over matched Direct, with larger gains for Qwen3-VL-4B.

Reasoning cost. RAVEL improves Whole accuracy on both backends while using 1.26× the input tokens of matched Direct for Qwen3-VL-4B and 1.29× for Astra. Despite similar input growth, cumulative inference time reaches 3.45× the baseline for Qwen3-VL-4B but 1.17× for Astra. Astra thus retains accuracy gains with smaller relative time overhead; runtime also depends on hardware and serving conditions.

6. Understanding Relational Reasoning with RAVEL

6.1. Relation Updates and Trajectory Consistency

Frequent preservation makes predicting no change a tempting shortcut. RAVEL improves Qwen3-VL-4B's changed joint accuracy from 2.29% to 52.00% under matched access while reducing false change alarms on both rechecked and unrelated relations. Changed recall also increases, showing that the model detects more real changes as it becomes less likely to invent them. This balance is backend-dependent: Molmo2-ER gains precision in the full-cohort comparison but loses some recall. Better preservation can therefore accompany either broader change detection or a more conservative prediction pattern.

Requiring an entire trajectory to be correct exposes another gap. On matched T5, Qwen3-VL-4B establishes the correct initial relation vector in 60% of cases. Of the 100

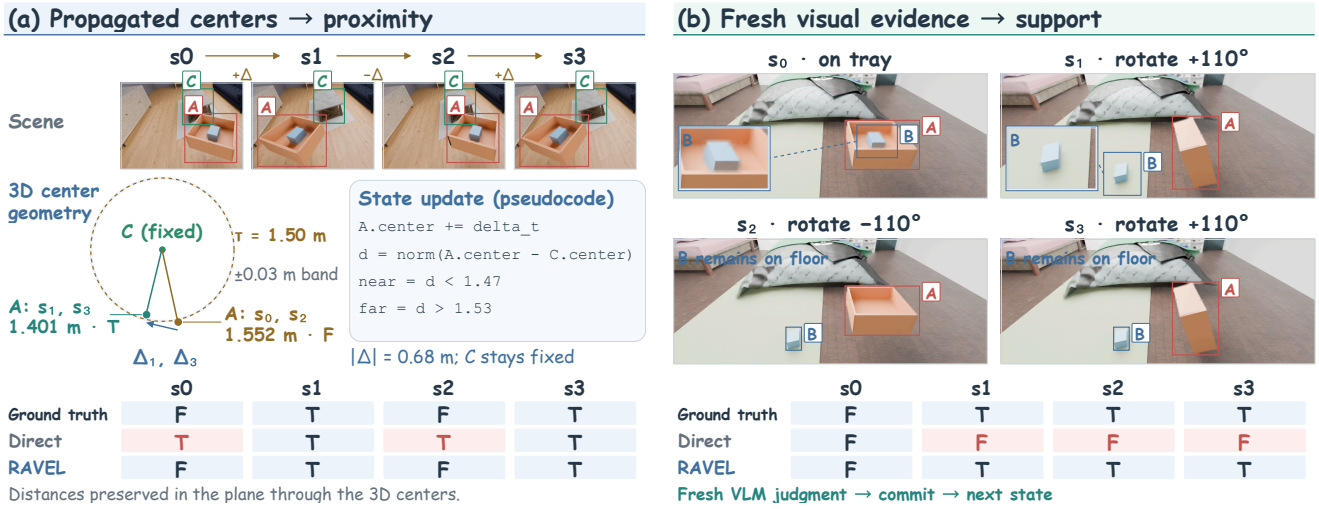


Figure 5. **Spatial computation and visual judgment across four states.** (a) Translation updates A's center while C remains fixed. The geometric inset preserves 3D distances in the plane through their centers; the circle marks 1.50 m with a ±0.03 m boundary band. Executing these updates yields alternating proximity judgments. (b) Four room views share the same crop. Insets at s_0/s_1 show B's support changing from tray to floor; later views show the tray rotating while B stays on the floor. Unresolved support requires fresh visual judgment. GPT-6-Astra predictions are shown, with errors in red. Crops are enhanced for display; inference uses both original views.

evaluated trajectories, nine have every state vector correct, twelve have every update label correct, and eight satisfy both requirements. Whole requires every state value and update label to match the annotation; correct update labels can coexist with incorrect relation values, and vice versa. Astra initializes more reliably, but later errors likewise limit trajectory correctness and its matched Whole gain remains small. We trace whether these errors arise in new judgments or inherited findings.

6.2. Applicability and Truth of Evidence

Validity tracking determines whether a finding still applies; source correctness determines what reuse preserves. All scorable program findings agree with annotations in the full Qwen3-VL-4B/Astra runs, with source accuracies covering different query subsets. Under unchanged fixed dependencies, reuse preserves the original judgment, including its mistakes: 428 erroneous Qwen3-VL-4B sources produce 917 wrong reuses, versus 114 and 224 for Astra. Fresh assessment can correct an earlier error or overturn a correct finding. Correct-to-incorrect transitions outnumber corrections: 3,118 versus 965 for Qwen3-VL-4B and 394 versus 199 for Astra. These distinct paths explain why applicable premises alone do not ensure joint correctness. The traces exclude unaligned T4 branch findings.

6.3. Qualitative Cases across Four States

Prescribed translation lets RAVEL track alternating A–C proximity across four states, while Direct predicts true throughout (Fig. 5a). The geometric inset shows how executable center updates carry the relation across the proximity boundary, grounding each computed answer. Tray rotation changes B–floor support after the initial state; RAVEL detects this transition, whereas Direct remains false (Fig. 5b). The room views preserve the action context, while two contact insets expose the change in support; these visual judgments complement center propagation for proximity.

6.4. Spatial Computation and Model Judgment

The benefit of spatial code depends on what the available premises can determine. Under matched access, Qwen3-VL-4B's direction accuracy rises from 46.49% to 85.68%, while support accuracy declines slightly and collision Whole remains low. Direction can use propagated centers; support and contact need evidence about surfaces, and free-motion outcomes require prediction after numerical premises become invalid. The local backends also struggle on the unseen-outcome task T2, where future observations are unavailable. Astra's matched contact and proximity scores decline slightly as well. The remaining physical and visual judgments lie beyond what the current executable premises resolve.

7. Limitations

RAVEL assumes accurate initial geometry, supported actions, and four identified objects in synthetic indoor scenes. Appearance tests do not evaluate noisy declarations or a real perception front end; reuse can preserve initial errors. Matched controls cover two backends with unequal benefits

and additional token use; they identify joint representation-and-execution effects and conditional binding/propagation effects.

8. Conclusion

RAVEL couples persistent spatial representations with executable updates, validity checks, relation evaluation, and answer composition. Matched controls show that shared representation and execution benefit both backends, with conditional gains from binding and propagation. Applicable premises complement model judgments; joint consistency also requires correct observations.

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